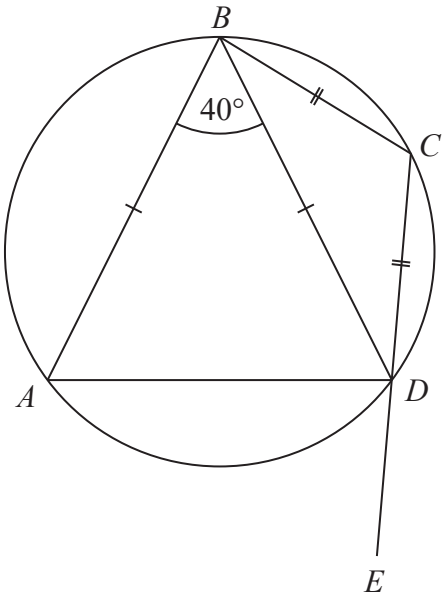
18 The points A, B, C and D lie on a circle.
CDE is a straight line.



$BA = BD$
$CB = CD$
Angle $ABD = 40°$

Work out the size of angle ADE.
You must give a reason for each stage of your working.

BAD = BDA = (180° – 40°)/2 = 70°

1 mark
70° in isosceles ABD.

Reason: base angles of isosceles triangle ABD.

angle BCD = 180° – 70° = 110°

1 mark
Opposite cyclic angles 180°.

Reason: opposite angles in cyclic ABCD.

angle CBD = angle BDC = (180° – 110°)/2 = 35°

Reason: base angles of isosceles triangle BCD.

angle ADE = 180° – (70° + 35°) = 75°

1 mark
ADE = 75°.

Reason: angles on straight line CDE.

2 marks total
Cyclic and other reasons linked.

(Total for Question 18 is 5 marks)